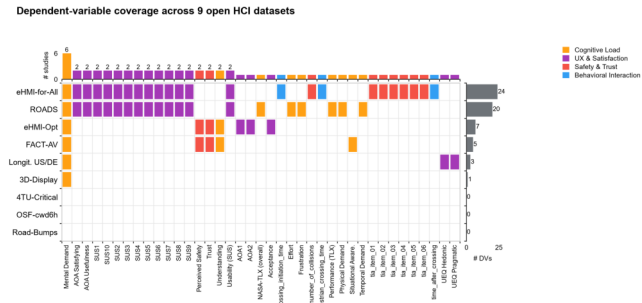


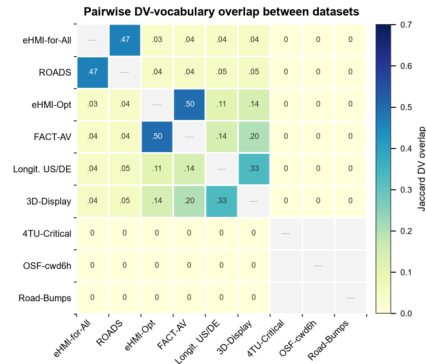
OpenDV-HCI: standardizing dependent variables turns fragmented open datasets into comparable, synthesizable evidence

(a) DV coverage across 9 datasets



After schema-driven standardization, mean pairwise DV overlap is only Jaccard = 0.067. Only Mental Demand (NASA-TLX) is shared by >2 studies; 3 sensorlog datasets expose no canonical questionnaire DV.

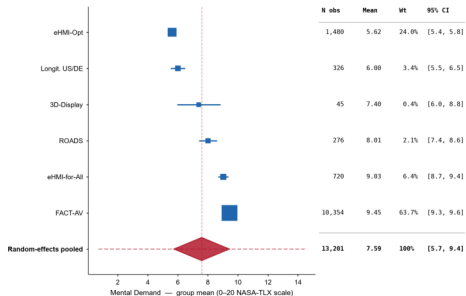
(b) Pairwise overlap (Jaccard)



(c) Meta-analysis: Mental Demand

Evidence synthesis unlocked by standardization: NASA-TLX Mental Demand

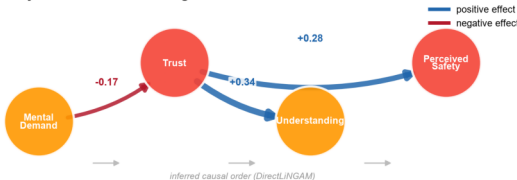
Pooling is only possible because heterogeneous source labels were mapped to one canonical DV. Very high P shows that shared names still mask scale/population differences.



Random effects (DerSimonian-Laird), k = 6 studies. Heterogeneity: $I^2 = 99.6\%$, $\tau^2 = 5.23$, $Q(5) = 1394$, $p < 0.001$

(d) Cross-study causal structure

Cross-study causal structure among standardized DVs



DirectLINGAM on pooled within-study z-scores (complete-case, non-synthetic; $n = 11,094$ obs). Edge weights are structural coefficients.